# DANIEL CHAHINE

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PhD candidate in Biomedical Engineering at UTS, researching monolithic scintillator detectors for PET through Monte Carlo simulation, information theory, machine learning and image reconstruction. First Class Honours graduate whose capstone built a CUDA-accelerated, continuous-coordinate OSEM reconstruction pipeline. Second author in IEEE JBHI; delivered a quality-control system into commercial manufacturing for Cochlear.

## Education

**PhD, Biomedical Engineering** Jul 2026 – Present

*University of Technology Sydney | Supervised by Daniel Franklin, co-supervised by Roumani Alabd*

- Research: Monte Carlo optical simulation and information-theoretic bounds for PET detectors.

**Bachelor of Engineering (Honours), Biomedical Engineering** Mar 2022 – Jul 2026

*University of Technology Sydney* | First Class Honours

- GPA 6.72/7 | WAM 88.6 | Dean's List 2023–2026.
- Capstone: CUDA-accelerated list-mode OSEM reconstructing PET images directly from continuous 3D LOR endpoints, with no sinogram binning step, validated on simulated point sources and a NEMA IEC phantom.

## Publications

Sun, Y., **Chahine, D.**, et al. (2025). *Voxel-Level Brain States Prediction Using Swin Transformer.* IEEE Journal of Biomedical and Health Informatics. Selected for oral presentation at IEEE BHI 2025.

Abraham, R., **Chahine, D.**, Franklin, D.R. (2026). *PRISM: A Physics-Informed Spatiotemporal CNN for Event Localisation in Nanocomposite Monolithic PET Scintillators.* Physics in Medicine & Biology. *(Under Review)*

## Research & Technical Experience

**Computational Biologist** | Garvan Institute Summer Scholarship Nov 2025 – Feb 2026

*Susan Clark Lab, supervised by Dr. Ruth Pidsley*

- Developed epigenetic analysis pipeline processing ChIP-seq data (4 histone marks) across 62 paired CAF/NPF fibroblast cell lines and 2 epithelial cell lines to analyse prostate cancer risk variants.
- Implemented ChromHMM to segment chromatin states genome-wide; filtered 3,700 GWAS SNPs to identify 146 variants in fibroblast-specific regulatory regions.
- Integrated PCHiC and RNA-seq data to link enhancer variants to target genes (Python, Polars).

**Research Engineer** | University of Sydney & Brain and Mind Centre Dec 2024 – Aug 2025

**Data Engineer** Jun – Aug 2025

*Supervised by Prof. Wei Chen, Dr. Jia Liu, and Dr. Shawn Dexiao Kong*

- Built EEG preprocessing pipeline for AI-based sleep spindle segmentation, using wavelet-transform frequency-domain features; prototyped Swin Transformer and U-Net segmentation models.

**Biomedical AI Researcher — Engineering Accelerator** Feb – Jun 2025

*Supervised by Prof. Fernando Calamante and Prof. Jinglei Lv*

- Extended 4D Swin Transformer U-Net for brain state forecasting, improving prediction stability from 10 to 200 frames using autoregressive inference.
- Implemented a combined SSIM and masked MSE loss for long-horizon stability; led model validation across DGX and A100 clusters.

**Biomedical AI Researcher — Vacation Scholarship** Dec 2024 – Feb 2025

*Supervised by Prof. Fernando Calamante and Prof. Jinglei Lv*

- Developed novel 4D Swin Transformer U-Net architecture for voxel-level brain state prediction using resting-state fMRI, achieving a masked MSE of 0.035 on held-out test subjects.
- Optimised model through skip connections and autoencoder architecture connectivity improvements; leveraged USYD RDS and BMC NVIDIA DGX Server.

**Society Executive** | UTS Biomedical Engineering Society May 2024 – 2026

**Chief Technical Officer** Jan 2025 – 2026

- Architected system integration for humanoid robot: designed component interfaces, communication protocols (high-bandwidth serial), and software distribution across embedded and host systems.
- Managed team of 20; $2,000 budget, delivering 2 projects (computer vision-controlled robotic hand, humanoid robot).
- Showcased to 500+ attendees across 5 industry networking events and UTS TechFest event.

**Project Software Lead** May – Aug 2024

- Led 4-person software team developing biomimetic robotic hand controlled by computer vision.
- Architected control software integrating MediaPipe joint detection, vector calculation algorithms, and servo control mapping.

**Quantum Computing Researcher** | UNSW Sydney Quantum Academy Jan – Feb 2024

*Supervised by Dr. Maja Cassidy*

- Contributed to electron-on-neon quantum computer development and modelling.
- Designed and manufactured $3,000 quantum chip housing; reduced seal count from 20+ to 4 (fewer points of failure) and expanded signal-carrying capacity by over 25% with multiaxial cabling.
- Performed RF simulations (Ansys HFSS, Q3D); developed automation scripts in QCoDeS and LabOne Q.

**Engineering Consultant** | OPTIK Consultancy (Client: Cochlear) May – Aug 2023

*Supervised by Osman Mah*

- Designed automated QC system for implantable medical devices; prototyped across 6 devices, achieving >95% classification accuracy.
- Replaced manual X-ray inspection with <2 second automated assessment; system entered commercial manufacturing after handover.
- Implemented FMEA, CAPA, and DMR documentation; identified critical memory leak in vendor API through stress testing.

## Awards & Funding

- **ActivateUTS Award for Most Innovative Student Initiative** (2026) and **ActivateUTS Strategic Project Grant** (2026, $1,000), for the society's humanoid robot.
- **Research scholarships:** University of Sydney Vacation Research ($4,000), Engineering Accelerator ($1,500) and Winter Vacation Research ($3,000), 2024–2025; Sydney Quantum Academy ($3,333), 2024; Garvan Institute Summer Scholarship ($6,250), 2025–2026.

## Technical Skills

**Languages:** Python, C/C++, Bash, MATLAB, SQL, Rust, Java
**ML/DL:** PyTorch, CUDA | Transformers, U-Nets, PointNets, RNNs/LSTMs, Physics-Informed NNs
**Computer Vision:** MediaPipe, YOLO, OCR | Medical imaging (fMRI, EEG, PET)
**Bioinformatics:** ChIP-seq, RNA-seq, PCHiC, ChromHMM, Polars
**Simulation & Quantum:** Monte Carlo (GATE, Geant4, Celeritas), Ansys HFSS/Q3D | PennyLane, Qiskit, QCoDeS
**Infrastructure:** Linux, HPC clusters (DGX, A100), AWS/Lambda, Git, Docker
**Open source:** Celeritas contributor (ORNL GPU transport): fixed an optical-scattering physics bug (PR #2648).
**Engineering:** SolidWorks, Fusion 360, KiCad, LTspice | Embedded (ESP32, Arduino)